Proving Data Structures Correct: Balanced Search Trees

Computer-Assisted Theorem Proving (7CCSACTP)

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Structural Invariant `complete`

All leaves are at the same level:

```
complete Leaf = True
complete ⟨l, _, r⟩ = (h(l) = h(r) ∧ complete l ∧ complete r)
complete ⟨l, _, m, _, r⟩ =
  (h(l) = h(m) ∧ h(m) = h(r) ∧
   complete l ∧ complete m ∧ complete r)
```

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Complete Trees Have Logarithmic Height

Lemma

$\texttt{complete t} \implies 2^{h(t)} \leq |t| + 1$

Proof by induction on `t`:

- `Leaf`: $2^0 = 1 = |\texttt{Leaf}| + 1$
- $t = \langle l, a, r \rangle$: by `complete`, $h(l) = h(r) = k$, so $h(t) = k + 1$. By the IHs for l and r: $2^{k+1} = 2^k + 2^k \leq (|l| + 1) + (|r| + 1) = |t| + 1$
- $t = \langle l, a, m, b, r \rangle$: $h(l) = h(m) = h(r) = k$, and $|t| = |l| + |m| + |r| + 1$. By the IHs for l and m: $2^{k+1} = 2^k + 2^k \leq (|l| + 1) + (|m| + 1) \leq |t| + 1$

$\Rightarrow h(t) \leq \log_2(|t| + 1)$: `isin` takes time $O(\log |t|)$

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